
Decision on submission to Journal of Quantitative Spectroscopy and Radiative Transfer

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Manuscript Number: JQSRT-D-26-00214

Block-Krylov discrete-dipole and volume-integral-equation solvers for multi-orientation light scattering across the dielectric-metal contrast range

Dear Dr. Moteki,

Thank you for submitting your manuscript to Journal of Quantitative Spectroscopy and Radiative Transfer.

I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following major revision. I invite you to resubmit your manuscript after addressing the comments below. Please resubmit your revised manuscript by **Jun 29, 2026**.

When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed. Please note that your revised submission may need to be re-reviewed.

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Journal of Quantitative Spectroscopy and Radiative Transfer values your contribution and I look forward to receiving your revised manuscript.

Kind regards,
Miroslav Kocifaj
Editor-in-Chief

Journal of Quantitative Spectroscopy and Radiative Transfer

Editor and Reviewer comments:

Reviewer #1: This is an excellent and technically mature manuscript. The author presents two paired open-source volume-integral light-scattering solvers, block-DDA Py and block-VIEM.jl, designed to accelerate multi-orientation forward simulations needed for simulation-based inference of particle properties from CAS-v2-type optical measurements. The core idea of amortizing many orientations through block-Krylov subspace methods is both elegant and practically important.

The manuscript is strong in several respects. First, the motivation is clear: existing general-purpose DDA and volume-integral solvers can represent arbitrary particle shapes and compositions, but orientation sweeps remain computationally expensive. Second, the theoretical formulation is careful, especially the fixed-particle/many-incident-direction transformation and the block linear-system representation. Third, the benchmark design is broad and convincing, spanning spheres, oblate spheroids, sphere doublets, and Gaussian random spheres across dielectric and high-contrast materials. Finally, the comparison against Mie theory, EBCM, MSTM, and Richardson extrapolation gives the reader confidence in the numerical conclusions.

The results are also useful for the wider light-scattering community. The contrast-dependent solver-selection rule, namely DDA for optically softer dielectric particles and VIEM for strongly absorbing or high-contrast particles, is clearly supported by the benchmark evidence. The monotonic per-orientation speedup with block size up to $L = 128$ is a particularly important practical result, and the discussion of near-singular modes and block-GMRES behavior provides a plausible physical and numerical explanation.

I recommend publication after minor revision. My comments below are intended to improve clarity, contextualization, and reproducibility rather than to question the main findings.

1. The Introduction would benefit from a slightly broader discussion of prior DDA work on aerosol aggregates and discretization accuracy, especially studies that tested DDA against MSTM for black-carbon aggregates and analyzed DDA resolution requirements (e.g., <https://doi.org/10.1016/j.jqsrt.2018.10.021> and <https://doi.org/10.5194/acp-19-2917-2019>). This would help readers see how the present block-Krylov acceleration complements earlier accuracy studies.

2. The manuscript states that DDA is appropriate for low-contrast targets and VIEM for high-contrast targets. This is well supported by the presented tests, but the authors may wish to clarify whether the suggested contrast boundary $|mp/mm|$ around 2 should be viewed as a practical

empirical guideline for the present discretization and tolerances, rather than a strict theoretical limit, because DDA performance is highly sensitive to refractive indices (see DOI10.1364/OE.20.016763).

3. Because the paper emphasizes open-source release, please ensure that the code repositories include a stable tagged version, installation instructions, example input files, and scripts or notebooks sufficient to reproduce at least a subset of the benchmark figures and tables.
4. For the Au cases, the use of a relaxed residual tolerance is reasonable, but the manuscript should explicitly state how much this relaxation affects the reported optical observables compared with discretization error, especially in the benchmark slots where exact references exist.
5. The benchmark uses Au as a demanding high-contrast stress test. Since the intended environmental applications include black carbon and iron oxides, it would be helpful to add one sentence explaining how the Au tests bracket or exceed the contrast challenge posed by those realistic aerosol materials.
6. Table C.3 is useful for practitioners. Please consider adding a sentence that identifies which memory component dominates for block-DDA Py at large L and which component dominates for block-VIEM.jl, since this will help users estimate workstation requirements.
7. Please check the consistency of notation for refractive index ratios, residual norms, and block size throughout the main text, figure captions, and appendices. The notation is mostly clear, but the paper is notation-heavy and would benefit from a final editorial pass.

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